

Mixture Models

Problem : Two-component Gaussian mixture models fitting by EM algorithm

- Assume a two-component normal mixture model for the following twenty fictitious data points and apply the EM algorithm to find the MLEs of the model parameters.
- -0.39, 0.12, 0.94, 1.67, 1.76, 2.44, 3.72, 4.28, 4.92, 5.53, 0.06, 0.48, 1.01, 1.68, 1.80, 3.25, 4.12, 4.60, 5.28, 6.22
- Data: Table 8.1 of McLachlan, G. and Peel, D. (2000): *Finite Mixture Models*, Wiley, N.Y.

```
y <- c(-0.39, 0.12, 0.94, 1.67, 1.76, 2.44, 3.72, 4.28, 4.92, 5.53,  
       0.06, 0.48, 1.01, 1.68, 1.80, 3.25, 4.12, 4.60, 5.28, 6.22)  
n <- length(y)
```

```
hist(y, breaks = 8, main="Twenty fictitious data")
```

```
# initial values
```

```
pi1 <- 0.5
```

```
pi2 <- 0.5
```

```
mu1 <- 1.0
```

```
mu2 <- 3.0
```

```
sigma1 <- 0.8
```

```
sigma2 <- 0.8
```

```
tiny <- 10^(-7) # a small value to test convergence
```

```
loglik.old <- 0
```

```
mix.pdf <- pi1*dnorm(y,mu1,sigma1) + pi2*dnorm(y,mu2,sigma2)
```

```
loglik.new <- sum(log(mix.pdf))
```

```
k<-1
```

```
# loop for EM-iterations
```

```
while(abs(loglik.new - loglik.old) >= tiny){
```

```
# E step
```

```
z1j <- pi1*dnorm(y,mu1,sigma1)/(pi1*dnorm(y,mu1,sigma1)+pi2*dnorm(y,mu2,sigma2))
```

```
z2j <- pi2*dnorm(y,mu2,sigma2)/(pi1*dnorm(y,mu1,sigma1)+pi2*dnorm(y,mu2,sigma2))
```

```
# M step
```

```
pi1 <- sum(z1j)/n
```

```
pi2 <- sum(z2j)/n
```

```
mu1 <- sum(z1j*y)/sum(z1j)
```

```
mu2 <- sum(z2j*y)/sum(z2j)
```

```

sigma1 <- sqrt(sum(z1j*(y-mu1)^2)/sum(z1j))
sigma2 <- sqrt(sum(z2j*(y-mu2)^2)/sum(z2j))

# log-likelihood
loglik.old <- loglik.new
mix.pdf <- pi1*dnorm(y,mu1,sigma1) + pi2*dnorm(y,mu2,sigma2)
loglik.new <- sum(log(mix.pdf))
k<-k+1
cat(c(k, pi1, pi2, mu1, mu2, sigma1, sigma2, loglik.new), '\n')
}
#-----

# Plotting fitted density
h <- hist(y, breaks=8, main="Twenty fictitious data")
# Add a Normal Curve
ynew <- seq(min(y),max(y),0.1)
obs.fit <- dnorm(ynew, mean=mean(y),sd=sd(y))
obs.fit <- obs.fit*diff(h$mids[1:2])*length(y)
lines(ynew, obs.fit, col="blue", lty=1, lwd=2)
# 2-fold Mixture density
mix.fit = pi1*dnorm(ynew, mu1, sigma1) +
          pi2*dnorm(ynew, mu2, sigma2)
mix.fit <- mix.fit*diff(h$mids[1:2]) * length(y)
lines(ynew, mix.fit, col="red", lty=2, lwd=2)
legend("topright", c("Without mixture density", "2-fold Normal mixture density"),
col=c("blue","red"), text.col=c("blue","red"), lty=c(1,2), lwd=2)
# -----

# Comparison with mixtools package-results
library(mixtools)
gm <- normalmixEM(y, k=2, lambda=c(0.5,0.5), mu=c(1.0,3.0),sigma=c(0.8,0.8))
gm$lambda
gm$mu
gm$sigma
gm$loglik
plot(gm)
#-----

# Plotting
plot(gm, which=2)
lines(density(y), col="black", lty=2, lwd=2)
# 2-fold Mixture density
mix.fit = gm$lambda[1]*dnorm(ynew, gm$mu[1], gm$sigma[1]) +
          gm$lambda[2]*dnorm(ynew, gm$mu[2], gm$sigma[2])
lines(ynew, mix.fit, col="blue", lty=3, lwd=2)
legend("topright", c("Without mixture density", "2-fold Normal mixture density"),
col=c("black","blue"), text.col=c("black","blue"), lty=c(2,3), lwd=2)
#-----

```

Problem : Generate three-component Gaussian mixture data and use the generated data set for estimating the parameters of the model by EM algorithm

```
# Generate three-component Gaussian mixture data
N <- 100000
# Sample N random uniforms U
U <- runif(N)
# gen.y Variable to store the samples from the mixture distribution
gen.y <- rep(NA, N)
# True parameters
p <- c(0.3, 0.5, 0.2)
m <- c(0, 10, 3)
s <- c(1, 1, 0.1)

#Sampling from the mixture
for(i in 1:N){
  if(U[i] < p[1]){
    gen.y[i] <- rnorm(1,m[1],s[1])
  }else if(U[i] < (p[1]+p[2])){
    gen.y[i] = rnorm(1,m[2],s[2])
  }else{
    gen.y[i] = rnorm(1,m[3],s[3])
  }
}

library(mixtools)
gm3 <- normalmixEM(gen.y, k=3, lambda=c(p), mu=c(m),sigma=c(s))
gm3$lambda
gm3$mu
gm3$sigma
gm3$loglik

#Plotting the true and estimated densities as for check
x <- seq(min(gen.y), max(gen.y), 0.1)
true.pdf <- p[1]*dnorm(x,m[1],s[1]) + p[2]*dnorm(x,m[2],s[2]) + p[3]*dnorm(x,m[3],s[3])
est.pdf <- gm3$lambda[1]*dnorm(x,gm3$mu[1],gm3$sigma[1]) +
  gm3$lambda[2]*dnorm(x,gm3$mu[2],gm3$sigma[2]) +
  gm3$lambda[3]*dnorm(x,gm3$mu[3],gm3$sigma[3])

plot(x, true.pdf, main="Density Estimate of the Mixture Model", col="black", lwd=2) lines(x,
est.pdf, col="red", lwd=2)
legend("topleft",c("True Density","Estimated Density"),col=c("black","red"),lwd=2)
#-----
```

Problem : Generate multivariate Gaussian mixture data and use the generated data set for estimating the parameters of the model by EM algorithm

- n : No. of observations
- k : No. of mixture components
- d : dimension of density
- μ : $k \times d$ matrix with means
- sig : $k \times d \times d$ matrix with $d \times d$ covariate matrices

```
gen.mix <- function(n, k, mu, sig) {
  library(MASS)

  d <- length(mu[1,])      # number of dimensions
  result <- matrix(rep(NA,n*d), ncol=d)
  colnames(result) <- paste0("X",1:d)

  for(i in 1:n) {
    result[i,] <- mvrnorm(1, mu = mu[k[i],], Sigma=sig[,k[i]])
  }
  result
}
# -----

set.seed(101)
n <- 360

mu <- matrix(c(4.0,4.0, 5.0,5.0, 6.5, 5), ncol=2, byrow=T)

sigs <- array(rep(NA,2*2*3), c(2,2,3)) # 3D matrix
sigs[,1] <- matrix(c(.25, .21, .21,.25), nrow=2, byrow=TRUE)
sigs[,2] <- matrix(c(.25,-.21,-.21,.25), nrow=2, byrow=TRUE)
sigs[,3] <- matrix(c(.25, .21, .21,.25), nrow=2, byrow=TRUE)

pi <- c(.2,.5,.3) # mixing coeffs
classes <- sample(1:3, n, replace=TRUE, prob=pi)

mydata <- gen.mix(n, classes, mu, sigs)

# mydata <- read.csv("mydata.csv", as.is=TRUE)

plot(mydata, col=c("red","green","blue")[classes], xlab="X1", ylab="X2", pch=19)

library(mixtools)
model <- mvnnormalmixEM(mydata, k=3, epsilon=1e-04)
model$mu
model$sigma
plot(model, which=2)
# -----
```

Problem : The data given in the mvn3c2dmix.csv file follow a multivariate Gaussian mixture distribution with 3 components in 2 dimensions.

- a) Plot the data with respect to classes and comment on the plot
- b) Estimate the parameters of the model using EM algorithm and comment on the estimates
- c) Draw the estimated density curves for the mixture model and comment.

Solution

```
mydata <- read.csv("mvn3c2dmix.csv ", as.is=TRUE)
mydata

plot(mydata, col=c("red","green","blue")[classes], xlab="X1", ylab="X2", pch=19)

library(mixtools)
model <- mvnnormalmixEM(mydata, k=3, epsilon=1e-04)
model$mu
model$sigma
pi <- model$lambda
plot(model, which=2)
# -----
```